DURGA SANDEEP NAMBALLA

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PROFESSIONAL SUMMARY

Game Developer with 9 years of industry experience in Unity3D (C#), building 2D/3D mobile games and gameplay systems. Recently completed my Master's degree in Computer Science, specializing in Computer Graphics Programming. Brings teaching experience with strong mentoring and communication skills.

EDUCATION

DigiPen Institute of Technology — Seattle, USA
Master of Science in Computer Science — 2024 - 2026

Jawaharlal Nehru Architecture and Fine Arts University — Hyderabad, India
Bachelor of Science (Computer Science and Game Development) — 2011 - 2015

PROFESSIONAL EXPERIENCE

Backstage Pass Institute of Gaming and Technology — Jan 2019 – Aug 2024

Assistant Professor — Hyderabad, India

- Mentored students in coursework and an incubator program, guiding projects from concept to product development.
- Taught game development using Unity3D and C#, enabling students to design and build interactive projects.
- Integrated technology into instruction and developed course materials to enhance engagement, critical thinking, and problem-solving.
- Provided individualized support, assessed performance, and delivered constructive feedback to improve student outcomes.

TapNext Studios — Sep 2017 – Sep 2018

Game Developer — Chennai, India

- Developed 5+ 2D and 3D mobile games for Android using Unity3D and C#, collaborating within a 5-member cross-functional team to ship production-ready titles.
- Designed and customized scalable game systems for 2+ sister companies, adapting to diverse product requirements and reducing integration time by 30% across projects.
- Built reusable gameplay frameworks and modular systems incorporating core gameplay mechanics and AI-driven behaviors, decreasing development time for new features by 30%.
- Managed the end-to-end game development pipeline from concept to production, delivering multiple titles on schedule and reducing production bottlenecks by 20%.

Nextwave Multimedia — Aug 2016 – Sep 2017

Game Developer — Hyderabad, India

Project: World Table Tennis Champs — [Demo]

- Contributed to the development of a mobile 3D Table Tennis game from early concept through pre-launch using the Unity game engine, optimizing performance for resource-constrained devices.

- Implemented adaptive AI opponent behaviors with multiple difficulty levels, adjusting to player gameplay patterns to improve challenge and engagement.
- Built core gameplay modules, including tournament data generation and dynamic menu systems, applying Singleton and Object Pooling patterns to reduce memory overhead.
- Architected UI update systems using the MVC pattern to enforce separation of concerns and improve code maintainability, collaborating with design and art teams on a production-ready build.

SuperNova Games Oct 2015 – Sep 2016

Game Developer Hyderabad, India

Project: Cricket Dash (3D Mobile Infinite Runner) [Demo]

- Developed a 3D mobile infinite runner game using Unity3D and C#, serving as the sole programmer in a 2-member team alongside an artist and designer.
- Implemented core gameplay systems, including game logic, player mechanics, and progression systems.
- Leveraged object pooling to reduce memory overhead and maintain smooth real-time gameplay on mobile devices under increasing game pace.
- Architected a data-driven achievement system using JSON for configurable definitions and C# events/delegates to enable event-based tracking and real-time updates.

ACADEMIC PROJECTS

Course Projects – Computer Graphics & Game Development

• Advanced Computer Graphics (OpenGL, C++) [Github]

- Extended a prebuilt C++/OpenGL graphics framework, implementing the core 3D rendering pipeline including homogeneous coordinate transformations and perspective projection.
- Developed a physically based GLSL microfacet BRDF lighting model incorporating diffuse, specular, and Fresnel effects for realistic shading.
- Integrated key rendering techniques include texture mapping, sky dome rendering, normal mapping, and dual-paraboloid environment reflection mapping.
- Designed and optimized a two-pass shadow mapping pipeline using FBOs and shader-based techniques and built HDR image-based lighting (IBL) and irradiance mapping workflows to enhance visual realism.

• Advanced Real-Time Rendering (OpenGL, C++) [Github]

- Implemented a deferred shading pipeline in OpenGL supporting multiple dynamic local lights for scalable real-time rendering.
- Developed Moment Shadow Mapping (MSM) techniques for soft, high-quality shadows, and built PBR/HDR image-based lighting (IBL) pipelines with spherical-harmonic irradiance maps for realistic global illumination.
- Implemented screen-space ambient occlusion (SSAO) with bilateral filtering to improve depth perception and reduce noise artifacts.

• 2D Game Engine (C++, OpenGL) [Github]

- Collaborated within a 4-member team to build a custom game engine using C++ and OpenGL, designing UI systems, debugging tools, and contributing to the rendering pipeline.
- Developed and evaluated three gameplay prototypes, selecting the most viable concept and delivering a fully functional game.

• 3D Game Engine (C++, OpenGL) [Demo]

- Served as a Graphics Programmer and Producer in a 4-member team, designing and implementing core engine systems with a focus on rendering and graphics pipeline features.
- Built gameplay systems leveraging engine capabilities to ensure seamless engine-game integration, conducting playtesting to iterate on performance and user experience.

Course Projects – Machine Learning and Deep Learning

- Machine Learning 1 – Project (Speech Emotion Recognition) [Github]
  Python, NumPy, Seaborn, Matplotlib
  - Evaluated Random Forest and SVM classifiers on 6-class emotion recognition (CREMA-D, 7,442 audio samples), using balanced accuracy and macro-AUC comparisons (SVM: 0.82 vs. RF: 0.80) to assess class discrimination under imbalance.
  - Optimized Random Forest efficiency through constrained tree depth, feature subsampling, and SelectFromModel selection, while analyzing scalability trade-offs against SVM for deployment.

- Machine Learning 2 – Project - Movie Recommender System – Collaborative Filtering, Matrix Factorization & Hybrid Integration [Github]
  Python, NumPy, Seaborn, Matplotlib
  - Co-developed a hybrid movie recommender combining cluster-based KNN collaborative filtering, SVD matrix factorization, and a knowledge-based recommender on MovieLens 100K (943 users, 1,682 movies, ~94-97% sparsity).
  - Applied K-means++ segmentation to cut KNN RMSE to 1.112 and tuned SVD to a best RMSE of 1.067, integrating both into a hybrid system with dynamic CF/KB weighting and fallback logic for guaranteed coverage.

- Deep Learning - Project – Convolutional Neural Networks for Image Classification (TensorFlow/Keras, Google Colab) [Github]
  - Designed and trained CNN architectures across CIFAR-10, CIFAR-100, and Animal-10 (~78% validation accuracy on CIFAR-10), and applied transfer learning with MobileNetV2 using a two-phase fine-tuning strategy.
  - Built a config-driven experiment pipeline with checkpointing and CSV/TensorBoard logging, using EarlyStopping, ReduceLROnPlateau, and class weighting to stabilize convergence under class imbalance.

- Deep Learning - Project – Deep Reinforcement Learning – DQN Vacuum Agent (PyTorch) [Github]
  - Built a Deep Q-Network from scratch in PyTorch, implementing experience replay, epsilon-greedy exploration, target networks, and custom reward shaping to train an agent to navigate and clean grid environments.
  - Scaled and trained the agent across 5x5, 10x10, and 15x15 grid environments, visualizing training performance and agent behavior via text and Pygame rendering.

SKILLS

- Languages: C++, C#, Python.
- Libraries: OpenGL, GLFW, GLAD, GLM, Assimp, FreeType.
- Game Engines: Unity (C#).
- Scripting & Tools: Lua, sol2, DOTween.
- Machine Learning & Deep Learning: scikit-learn, TensorFlow/Keras, PyTorch, Pandas, NumPy, Matplotlib, Seaborn